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Optimisation of Hydropower Operation in a Liberalised Market with Focus on Price Modelling

Birger Mo, Anders Gjelsvik, Asbjørn Grundt and Kjetil Kåresen

Abstract—**This paper describes the structure and identification of a price model that is used in stochastic optimization of hydro operation and flexible contracts. The price model must be simple in order to be applicable in a stochastic optimization framework and the model should incorporate as much of the statistics of the price process as possible. Modelling of extremes is an important factor for the simulation capabilities of the optimisation models. The paper shows examples of simulated optimal operation of hydropower plants with the new price model. The paper also shows how the price model is used in a model that integrates hydro operation and financial hedging. In the forward market, prices of contracts with delivery several years ahead vary from one week to the next. In order to model this long-term uncertainty we have amended our spot price model.**

I. Introduction

OPTIMAL operation of hydropower plants and use of flexible contracts in a liberalised market is very dependent on spot price forecasts. Such price forecasts can be made using several different methods. The market approach to price forecasting could assume that that the expected spot prices are equal to the current prices in the forward market. Uncertainty in future spot market prices would be derived from current options prices. In Scandinavia almost every utility also use a fundamental modelling approach to the price-forecasting problem by using the EMPS model [1]. This model describes supply and demand and gives as a result future balance prices for different futures of temperature and inflow. Thus, the price forecast will be given by a number of scenarios for the future. These scenarios also give a direct link between spot price and local inflow.

There are several causes why the fundamental price modelling approach is so popular in the Scandinavian market.

Firstly, the market prices are very dependent on physical variables which can be assumed to be known (e.g. snow pack reservoir) or variables for which there are reliable statistics (e.g. temperature and inflow).

Secondly, the physical production system, of which about 50 percent is hydro power, is very complex with many constraints and consists of nearly 1000 different hydro plants. However, technical information and rehabilitation schedules for each plant are known or can be obtained by the players. Reservoir content by region and the necessary physical data for detailed modelling are available.

Thirdly, the Scandinavian market consists of many players and the perfect market assumption of the EMPS model is a reasonable good assumption.

The market players with hydro generation and or end-user sales need to include the correlations between inflow, end-user sales and prices into their risk management systems. Low inflows to hydro reservoirs are positively correlated with high market prices and low temperatures (resulting in high end–user sales) are positively correlated with high market prices. These correlations cannot be estimated from current market prices (spot, futures and options). For hydro producers correlations between accumulated inflow for a whole season or year and market price are much more important than the correlations between weekly inflow and market price because of storage capacity. One to five years of relevant historical price data is not enough to estimate all relevant correlations.

Thus, the price forecast used by almost all utilities in Scandinavia has its origin in the EMPS model. The forecast consists of a number of possible spot price scenarios for the future, each scenario is assumed to have equal probability. The scenarios are constructed based on historical years, e.g. the sequence 1931-1933 gives scenario number one and 1932 to 1934 gives scenarios number two etc. assuming a three year planning horizon. The historical year give inflow to each reservoir and temperature and thus load for each area in Scandinavia. Since there are only 70 years of complete historical data (1931-2000) we will at most get 70 future price scenarios from the EMPS model.

The scenarios may in some cases be modified to actual market conditions. For example, the scenarios may be adjusted in order to get the same expected future spot price as the current prices in the forward market. An example of such a price forecast is shown in Fig. 1. The forecast consists of 60 scenarios. The forecasting horizon is two years with weekly time resolution.

B. Mo and A. Gjelsvik are with SINTEF Energy Research (e-mail: birger.mo@energy.sintef.no and anders.gjelsvik@energy.sintef.no).
A. Grundt is with Statkraft SF (e-mail: asbjorn.grundt@statkraft.no)
K. Kårensen is with Norwegian Computing Center (e-mail: Kjetil.Kaaresen@nr.no)

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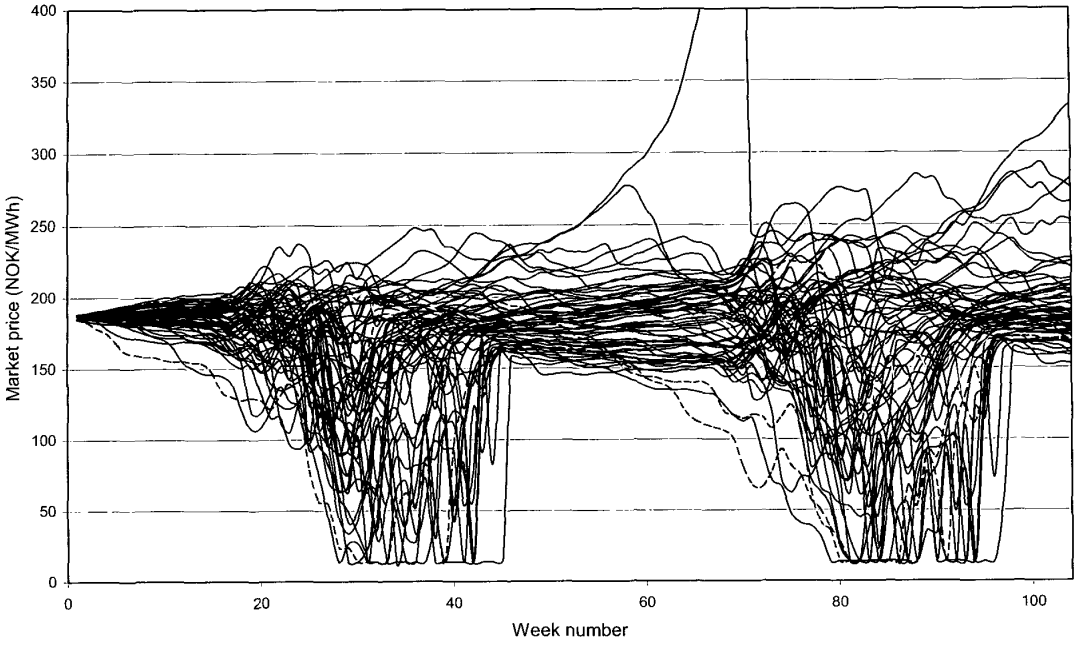


Fig 1. An example of price scenarios from the EMPS model, the highest price that is not included in the figure is 700 NOK/MWh

The price forecast has some important properties:

- The price in one week is strongly correlated with the price in the previous week.
- There is a given minimum price, for example zero or given by the cost of hydro generation.
- The price forecast has mean reversion properties.
- There are seasonal variations.

In the present paper we shall deal with the construction of price models for use in hydro scheduling and/or financial hedging. In [2], [3] and [4] the hydro-thermal scheduling problem is formulated as a stochastic dynamic optimization problem with weekly time resolution. The goal of these models is twofold:

- Find the optimal first stage decision
- Simulate/forecast optimal operation and profit for the future

To do this we need to include in the problem description a stochastic model of the future spot prices that can be used in a stochastic dynamic optimization framework, in our case based on dynamic programming principles. The price model is the same whether it is used in stochastic dynamic programming approach (SDP) or in a stochastic dual dynamic programming approach (SDDP). This model should be based on a price forecast such as in Fig. 1 and contain as much of the statistics as possible. In [2], [3] and [4] a Markov-like price model was used, without detail explanation. In the present paper, we describe the detail of the model identification. At the same time the model identification has been improved in many details compared to [2], [3] and [4]. We also show results from the use of the models in scheduling.

II. Price Model

The general price model structure is shown in Fig. 2. The model is a discrete Markov chain. It consists of a given number of price nodes, $P_i(t)$, in each time step t and transition probabilities $p_{i,j}(t)$, where $p_{i,j}(t)$ is the probability that the price is $P_j(t+1)$ at time step $t+1$ given that it was $P_i(t)$ at time step t. Note that the chain is inhomogeneous in time. Both the price nodes and transition probabilities vary from one time step to the next. In Fig. 2 there are five price nodes in each time step. Each node represents a given spot price. The identification problem concerns how to identify what price each node should represent, and how to estimate the transition probabilities. The basis for the identification is the price scenarios as shown in Fig. 1.

The main principle used in establishing the price model is the following: At every week t, there is one price value of each scenario, that is, a total of *Ncen* values. If N price nodes are wanted we divide the price range in N non-overlapping intervals, which may differ in size. The number of price values in each interval may also differ. In each interval, we compute the mean price, and this is taken as the price of the corresponding price node.

To estimate the transition probabilities for price from one week to the next, we follow the price scenarios and register how price changes from one price interval to another when we go to the next week. The details are given below.

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A. Identification of Price Nodes

The first part of the identification process is to specify the number of price nodes and to identify which price each node should represent. Computation of the price model is implemented as a separate computer program where the user of the program specifies the number of price nodes (N). The user also specifies the minimum number of scenarios (N_{min}) necessary to define one node. The minimum number ensures that each transition probability is based a minimum number of possible transitions.

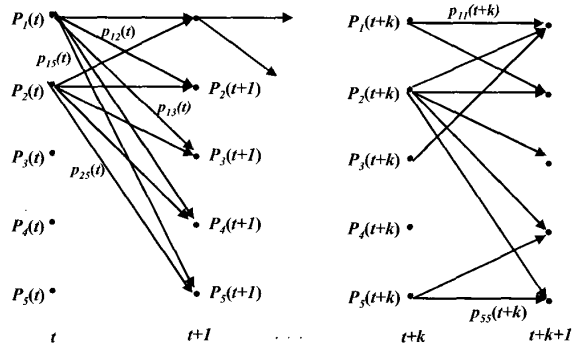


Fig. 2. Price model structure

The main principle of the node identification process is very simple. For given N_{min}, the lowest price node value for a given time step is given by the average of the N_{min} lowest scenarios (p_{min}) and correspondingly for the highest price node value (p_{max}). The price interval between p_{max} and p_{min} is divided into N–2 intervals of equal length and the price node values are given by the average of all the scenarios within each price interval. The actual implementation is somewhat more complex than described above. It could for instance be that there are more than N_{min} scenarios with the same minimum price, in this case all of these scenarios define p_{min}.

In could also be that some of the intermediate intervals do not contain N_{min} scenarios, the size of these intervals are increased until they cover N_{min} scenarios. The node identification is implemented in an iteration type algorithm where the first interval is placed with centre equal to average of all scenarios. The goal is to keep all intervals as equal in size as possible while ensuring that all intervals at least contain N_{min} scenarios.

The described identification is repeated independently for all time periods. Part of the node identification is to register which node each scenario belongs to in each time period. This information is used in estimation of the transition probabilities.

Testing has shown that if the total number of scenarios equals 70, five price nodes and minimum 6 scenarios to define a node usually give a 'good' result. If the available number of scenarios is larger the minimum should be increased.

B. Identification of Transition Probabilities

The second part of the identification is to quantify the transition probabilities between the nodes in time step t and t+1 for all t. The transition probabilities are found by solving a non-linear optimisation problem for each week. The goal of the optimisation is to maintain the conditional expected spot price one week ahead and the expected spot price seen from the start week. Define the following variables:

$N_i(t)$ The number of scenarios belonging to price node i in time step t.

$N^P_{i,j}(t)$ The number of scenarios belonging to node i in time step t and node j in time step $t+1$.

Initially the estimate of transition probabilities are computed simply as:

$$p_{i,j}(t) = \frac{N^P_{ij}(t)}{N_i(t)} \tag{1}$$

The next step is to modify the probabilities such that the price model has the following properties:

- Expected price in week t+1 given that the price in time step t is given by price node i should be equal to the average of those scenarios which belongs to node i in time step t.
- Assume that the correct price distribution is given for one time step. The expected future price should be equal to the average of all price scenarios for all future time steps.

The modification of the transition probabilities is formulated as an optimisation problem.

$$\min\left\{\sum_{i=1}^{N}\left(\sum_{j=1}^{N}\tilde{p}_{ij}(t)P_j(t+1)-E_i(t+1)\right)^2+\lambda\sum_{i=1}^{N}\sum_{j=1}^{N}(\tilde{p}_{ij}(t)-p_{ij}(t))^2\right\} \tag{2}$$

Subjected to the following constraints:

$$\sum_{j=1}^{N}\tilde{p}_{ij}(t)=1 \qquad \text{for all } i \tag{3}$$

$$\sum_{i=1}^{N}\tilde{p}_{ij}(t)N_i(t)=N_j(t+1) \qquad \text{for all } j \tag{4}$$

$$\tilde{p}_{ij}(t)\geq 0 \qquad \text{for all } i \text{ and } j \tag{5}$$

$$\tilde{p}_{ij}(t)\leq 1 \qquad \text{for all } i \text{ and } j \tag{6}$$

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Where

$\tilde{p}_{i,j}(t)$	Modified transition probabilities.
λ	A scaling factor, testing has shown that λ =0.01 is appropriate.
N	The number of price nodes.
$E_i(t+1)$	The average price in time step t+1 of those scenarios which belong to price node i in time step t.

λ is a balancing factor that also makes sure that the optimisation problem has only one solution. The problem is solved by routine E04UCF from the NAG Fortran library. The first part of the objective function regards the conditional expected price for the next time step and the second part is penalizing deviation from the original transition probabilities. Equation (6) is used to fulfill the requirement regarding the unconditional expected future price.

The optimization problem is solved for all time steps giving the new modified transition probabilities.

C. Including Extreme Prices in Simulations

In the basic model identification process the extreme prices in one time step is not necessarily included in the price nodes. For example a price node does not represent the price spike around week number 70 in Fig. 1, since it is only represented by one scenario. However, in the simulation part of the optimization models [2], [3] and [4], the calculated strategy is simulated for all future prices given by Fig. 1. It is therefore necessary to represent the extremes in the price model in order to get 'realistic' simulation results for the extreme scenarios also, which is very important for the users of the models. 'realistic' reservoir contents can be obtained from historical observations. The price model is therefore expanded with two additional price nodes for each time step, which represent the highest and the lowest price. Assume that the following matrix of transition probabilities describes the original price model:

$$O_1 = \begin{bmatrix} \tilde{p}_{11}(t) & \tilde{p}_{12}(t) & \tilde{p}_{13}(t) & \tilde{p}_{14}(t) & \tilde{p}_{15}(t) \\ \tilde{p}_{21}(t) & \tilde{p}_{22}(t) & \tilde{p}_{23}(t) & \tilde{p}_{24}(t) & \tilde{p}_{25}(t) \\ \tilde{p}_{31}(t) & \tilde{p}_{32}(t) & \tilde{p}_{33}(t) & \tilde{p}_{34}(t) & \tilde{p}_{35}(t) \\ \tilde{p}_{41}(t) & \tilde{p}_{42}(t) & \tilde{p}_{43}(t) & \tilde{p}_{44}(t) & \tilde{p}_{45}(t) \\ \tilde{p}_{51}(t) & \tilde{p}_{52}(t) & \tilde{p}_{53}(t) & \tilde{p}_{54}(t) & \tilde{p}_{55}(t) \end{bmatrix} \quad (7)$$

The new expanded price model, which includes the extremes, is described by the following matrix:

$$O_2 = \begin{bmatrix} \tilde{p}_{11}(t) & \tilde{p}_{12}(t) & \tilde{p}_{13}(t) & \tilde{p}_{14}(t) & \tilde{p}_{15}(t) & 0 & 0 \\ 0 & \tilde{p}_{11}(t) & \tilde{p}_{12}(t) & \tilde{p}_{13}(t) & \tilde{p}_{14}(t) & \tilde{p}_{15}(t) & 0 \\ 0 & \tilde{p}_{21}(t) & \tilde{p}_{22}(t) & \tilde{p}_{23}(t) & \tilde{p}_{24}(t) & \tilde{p}_{25}(t) & 0 \\ 0 & \tilde{p}_{31}(t) & \tilde{p}_{32}(t) & \tilde{p}_{33}(t) & \tilde{p}_{34}(t) & \tilde{p}_{35}(t) & 0 \\ 0 & \tilde{p}_{41}(t) & \tilde{p}_{42}(t) & \tilde{p}_{43}(t) & \tilde{p}_{44}(t) & \tilde{p}_{45}(t) & 0 \\ 0 & \tilde{p}_{51}(t) & \tilde{p}_{52}(t) & \tilde{p}_{53}(t) & \tilde{p}_{54}(t) & \tilde{p}_{55}(t) & 0 \\ 0 & 0 & \tilde{p}_{51}(t) & \tilde{p}_{52}(t) & \tilde{p}_{53}(t) & \tilde{p}_{54}(t) & \tilde{p}_{55}(t) \end{bmatrix} \quad (8)$$

The new transition probabilities are such that the statistical properties of the original price model are unchanged for all prices in between the original price nodes. This expansion does therefore only effect the simulation results for the extreme price scenarios. The first stage decisions are unchanged also.

III. Modelling Long-Term Price Uncertainty

Reference [4] describes a model for integrated risk management hydropower scheduling and contract management. Prices in the forward market are assumed to be equal to the conditional expected future spot price. In our opinion this is a reasonable assumption, but we shall not discuss its validity further.

The Markov model can thus be used to calculate future forward prices at all stages in the planning period. Since the price model in Fig. 2 is a first order model, the conditional expected price will be approximately equal to the unconditional price. This is a consequence of the well-known fact that a Markov model will tend to forget its initial state at an exponential rate. As a result the model will not include price uncertainty for forward contracts with delivery many time periods into the future.

If we look at a time series of observed prices of forward contracts with delivery for example 70 weeks ahead in the Nord Pool market, we will observe that these prices vary from one week to the next. That is, there is an observable uncertainty for contracts with delivery several years ahead that will not be included in the original price model when we use conditional expected future spot price as the forward price. The result of this is that our integrated model will not change future positions, e.g. one year ahead, which obviously is risky.

As an ad hoc solution to this problem, we have expanded the price model with new nodes and transition probabilities that models probability of shifts in forward prices.

Define the following variables:

$\Delta(t)$	Shift in spot price in period t (NOK/kWh)
$p_\Delta(t)$	Probability of shift $\Delta(t)$

The model for shift in futures prices is shown in Fig 3.

The shift $\Delta(t)$ in period t can be positive or negative. The probability of no shift is equal to 1-2 $p_\Delta(t)$. If the shift has

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occurred the probability of remaining in that state is 1.0. The shift model is superimposed on the original price model, thus giving a model three times as large as the original model. For every original price node there is a possibility of a positive or a negative shift as shown in Fig. 3. The price nodes in the new model consist of the original price nodes and new nodes which are given by the original ones plus/minus the shift $\Delta(t)$. The transition probabilities in the new model are given by combining the original ones and the probabilities in Fig 3. Because the shift model is symmetric, with expected value of zero, the expected price of the original price model is unchanged. For example the possible changes in the futures prices between time period t and t+1, for a contract with delivery two years into the future, will be given directly by Δ(t) and $p_\Delta(t)$.

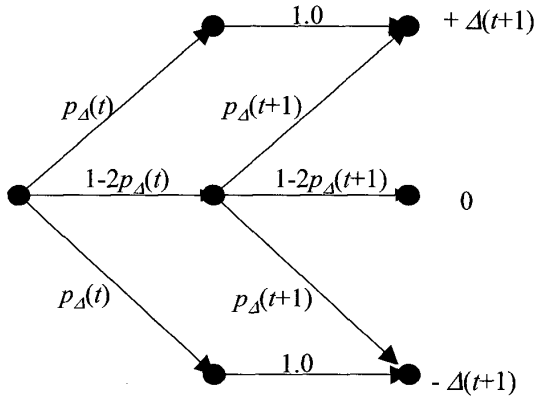


Fig. 3. Structure of model for shift in spot prices.

The shift $\Delta(t)$ and the probability $p_\Delta(t)$ are estimated from the observable uncertainty in forward prices. The difference between our original model and the new model is similar to the difference between one-factor and multi-factor models discussed in the literature, see e.g. [5]. The multi-factor models can describe the volatility of forward prices much better than the one-factor models.

The resulting effect is that the first stage hedging decisions in our new model will take into account price uncertainty for all forward contracts. The shift model is a relatively simple method to model uncertainty in long-term futures prices. However, the integrated model where this model is expanded model is used is very time consuming and it is essential that the number of price nodes is as small as possible, since the computation time is proportional to the number of price nodes. During simulation it is assumed for every stage that the shift has not happened. This is because the price scenarios do not include this information.

IV. Examples

The new price model has, as already mentioned, been implemented in several hydro scheduling models [2], [3] and [4]. The functionality described in section III is only used in the integrated scheduling and hedging model [4]. Norsk Hydro is currently testing this model in parallel with current risk management practice. The other models, reference [2] and [3], have been used with this new Markov model for about one year.

Fig. 4 shows simulated reservoir content for a given hydro system two years ahead for the same price scenarios as shown in Fig. 1. The optimal scheduling strategy is calculated using stochastic dynamic programming for a price model with and without modelling of extreme prices. The simulation results show that inclusion of extreme prices increase simulated minimum reservoir level during the winter period. Highest reservoir level for the second spring is also reduced. Highest reservoir level in the spring period for the first year is higher when extreme prices are included. When extreme prices are not included, highest simulated reservoir level in e.g. week 15 is caused by an extremely low price, when extremes are included highest simulated reservoir level in week number 15 corresponds to a high price.

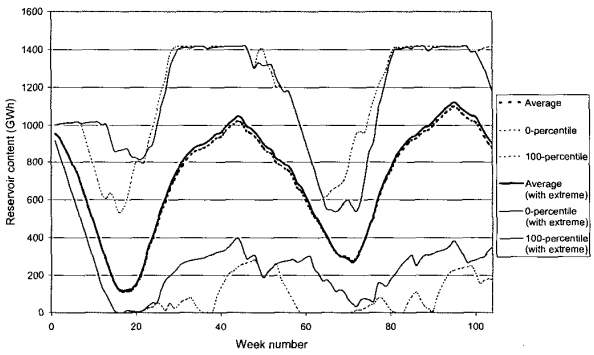


Fig. 4. Simulated reservoir content with and without modelling of extremes.

Fig. 5 shows conditional expected future spot price seen from week number 20 for all price nodes. Each line in Fig. 5 is labelled with the price (NOK/MWh) in week number 20, which is equal to the price node values. The figure shows as explained in the previous section that price for e.g. week number 104 is almost independent on the initial price, at least if the extreme price nodes is excluded. The extreme nodes have no influence on decisions in between the original price nodes.

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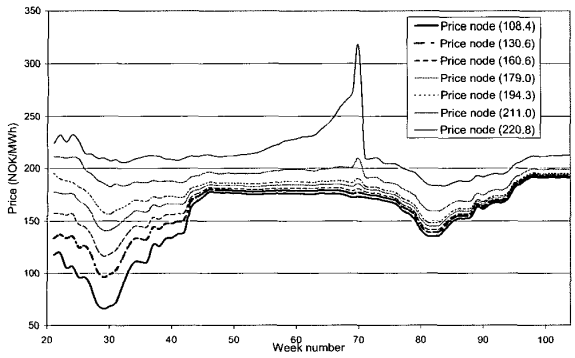


Fig. 5. Expected future spot price seen from the different price nodes in week number 20. Price node values (NOK/MWh) in week 20 are shown in brackets.

V. Further Work

The number of available price scenarios from the EMPS model (maximum 70) is too small for risk management purposes, ideally speaking. One utility has made their own model which can use the forecasts from EMPS model to generate several price scenarios for each inflow year. The model can in principle generate an infinite number of scenarios for each inflow year.

Another method with the same purpose is described in [6]. The approach is based on simulation of the EMPS model with observed input data. The difference between observed and simulated market prices is modelled by an autoregressive model (AR model). The AR model can be run in a forecasting mode and thus a large number of scenarios can be generated in addition to the original price scenarios from the EMPS model. This work was started recently, and many problems are still not solved.

VI. Conclusions

The paper has described a price model structure that is well suited to be used in a stochastic dynamic optimisation framework. The choice of model structure and time resolution is strongly influenced by the purpose of the optimisation, both first stage decisions and simulation. The paper has also showed how the model can be identified and some simulation results. The price model is implemented in several hydro-scheduling models. A large number of tests have verified that each improvement in model identification or model structure has improved the simulation results, i.e. improved economic results and more 'realistic' simulated reservoir content. Quantitative improvements in the first stage decisions are more difficult to verify except for the price shift modelling.

VII. Acknowledgment

We thank The Norwegian Research Council and Norsk Hydro who have sponsored parts of this work.

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IX. Biographies



Birger Mo received his M.Sc. in 1986 and a Ph.D. in 1991 in Engineering Cybernetics from the Norwegian Institute of Technology. He has since 1986 been employed at SINTEF Energy Research. His main interests are short-term forecasting, production planning and risk management.



Anders Gjelsvik received the M.Sc. in physics from the Norwegian Institute of Technology (NTH) in 1970. From NTH he also received a Ph.D. degree in engineering cybernetics in 1981. Since 1976 he has been with SINTEF Energy Research. His main interests are in large-scale optimization and numerical analysis.



Asbjørn Grundt received his M.Sc. in operations research in 1989 from the Norwegian Institute of Technology (NTH). In the period 1990 - 1994 he worked as a researcher at the Norwegian Research Defense Establishment. From 1994 to 2000 he worked in various positions in Norsk Hydro's Energy Division. Currently he is Vice President Operations, Statkraft SF's Market Division.

Kjetil Kåresen received the M.Sc. degree in statistics in 1992 and the Ph.D. degree in Applied and Industrial Mathematics in 1998, both from the University of Oslo. In this period he also worked for two years as an actuary in a Norwegian insurance company. From 1998 he has been working as a research scientist at the Norwegian Computing Center. His current research interests include statistical price modeling, statistical signal and image processing and general statistics.

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